NEC PAST Questions - Engineering Licence

Computer Engineering (Tribhuvan Vishwavidalaya)

SETI- NEC License Examination Mangshir,2080 - BCT

Section A (60*1 = 60)

1. Q: Which type of contract is not a "cost plus" contract?
a) Cost plus fixed fee b) Cost plus incentive fee c) Cost and fixed assets d) Cost plus percentage of cost
A: Cost and fixed assets

2. Q: In which year was the first expert system developed?
a) 1960 b) 1965 c) 1970 d) 1975
A: 1965

3. Q: What is the correct symbol for first angle projection?
a) ⊥ b) ∠ c) ⊿ d) ⊤
A: a)

4. Q: What is another name for a simple SR flip-flop?
a) A monostable multivibrator b) A bistable multivibrator c) An astable multivibrator d) A Schmitt trigger
A: A bistable multivibrator

5. Q: What is an alternative name for pipelining?
a) Assembly line operation b) Parallel processing c) Sequential execution d) Batch processing
A: Assembly line operation

6. Q: How many interrupts are there in the 8085 microprocessor?
a) 5 b) 8 c) 10 d) 13
A: 13

7. Q: Which of the following is not a non-maskable interrupt?
a) TRAP b) RST 7.5 c) RST 6.5 d) RST 5.5
A: Trap

8. Q: What is the relationship between terminal voltage and emf?
a) Terminal voltage is equal to emf b) Terminal voltage is greater than emf c) Terminal voltage is less than emf d) There is no relationship between terminal voltage and emf
A: Terminal voltage is less than emf

9. Q: If a delta connection has 3 ohms resistance, what is the equivalent resistance in star connection?
a) 1 ohm b) 3 ohms c) 9 ohms d) 27 ohms
A: 1 ohm

10. Q: What type of learning algorithm is Naive Bayes?

a) Unsupervised b) Supervised c) Reinforcement d) Semi-supervised
A: Supervised

11. Q: What are the input and output types for a 4:16 code converter decoder?
a) Input is binary and output is decimal b) Input is binary and output is hexadecimal c) Input is hexadecimal and output is binary d) Input is decimal and output is binary
A: Input is binary and output is hexadecimal

12. Q: Which logic gate is the building block for an encoder?
a) AND b) OR c) NOT d) XOR
A: OR

13. Q: Given the following code:

```
int *p, u;
float *y,x;
```

Which of the following is the correct next line of code?

a) p = &u b) p = u c) y = &x d) y = x

A: p = &u

14. Q: Which of the following is not a format specifier in C?
a) %d b) %f c) %s d) %t
A: %t

15. Q: In project management, what comes after the project idea in product design?
a) Development b) Testing c) Implementation d) Design
A: d) Design

16. Q: Which of the following is a stable filter?
a) RC filter b) LC filter c) Crystal oscillator d) Op-amp filter
A: Crystal oscillator

17. Q: How is frequency defined in AC?
a) Number of waves per second b) Time period of one wave c) Amplitude of the wave
d) Phase difference between voltage and current
A: Number of waves per second

18. Q: Who appoints the auditor of NEC?
a) NEC Executives b) Shareholders c) CEO d) Government

A: a)

19. Q: Which OSI layer is responsible for binary transmission?
a) Data Link layer b) Network layer c) Physical layer d) Transport layer
A: Physical layer

20. Q: What type of grammar is a PDA (Pushdown Automaton) associated with?
a) Type 0 grammar b) Type 1 grammar c) Type 2 grammar d) Type 3 grammar
A: Type 2 grammar

21. Q: What is a key characteristic of a Universal Turing Machine?
a) Programmable b) Fixed function c) Analog d) Quantum
A: Programmable

22. Q: Which test is used to determine machine intelligence?

a) Turing test b) IQ test c) A/B test d) Unit test
A: Turing test

23. Q: What component primarily uses BIOS?
a) CPU b) RAM c) Hard Drive d) Operating system
A: Operating system

24. Q: Real-time operation depends on what two factors?
a) Speed and accuracy b) Time and memory c) Input and output d) Hardware and software
A: a)

25. Q: What does distance vector routing depend on?
a) Hop counts b) Bandwidth c) Latency d) IP addresses
A: Hop counts

26. Q: What causes propagation delay?
a) Network congestion b) Router processing time c) Distance between routers d) Packet size
A: Distance between routers

27. Q: What is the maximum number of nodes a binary tree can have based on height h?
a) 2^h b) 2^h - 1 c) 2^(h+1) - 1 d) 2^(h-1)
A: 2^h - 1

28. Q: What is a requirement of a binary tree?
a) Must be balanced b) Must be complete c) Must be sorted d) Must have at least one child for each node
A: Must be sorted

29. Q: Which transformation resizes an object?

a) Translation b) Rotation c) Scaling d) Shearing
A: Scaling

30. Q: Which process runs after the Kernel bootstrap is executed?
a) /sbin/init b) /etc/init.d c) /boot/grub d) /proc/kmsg
A: /etc/init.d

31. Q: What is an encrypted message called?
a) Plaintext b) Ciphertext c) Keytext d) Hashtext
A: Ciphertext

32. Q: What is the function of the application layer in networking?
a) Data transmission b) Routing c) User to system application d) Error checking
A: User to system application

33. Q: On which plane does 2D rotation occur?
a) 3D plane b) 2D plane c) XY plane d) XZ plane
A: 2D plane

34. Q: Which of the following statements about functions and templates is incorrect?
a) Overloaded functions can be redefined b) Templates can be redefined c) Macros are efficient than templates d) All statements are correct
A: All statements are correct

35. Q: Which interrupt has the highest priority in 8086?
a) INTR b) NMI (Non-Maskable Interrupt) c) INT 0 d) INT 3
A: NMI (Non-Maskable Interrupt)

36. Q: How many 16-bit registers are there in the 8085 microprocessor?
a) 1 b) 2 c) 3 d) 4
A: 2

37. Q: What is the purpose of a Wait-for graph?
a) Process scheduling b) Memory allocation c) Deadlock detection d) File management
A: Deadlock detection

38. Q: What is the primary purpose of a special processor?
a) General computation b) Special function c) Data storage d) Network communication
A: Special function

39. Q: In networking, what term is used to describe excessive load?
a) Overload b) Congestion c) Saturation d) Bottleneck
A: Congestion

40. Q: What is a key feature of a bus in computer architecture?

a) Data transfer b) Priority handling c) Memory access d) Interrupt handling
A: Priority handling

41. Q: Which of the following is not a combinational circuit?
a) Multiplexer b) Decoder c) Adder d) Counter
A: Counter

42. Q: How many categories of memory storage are there?
a) 2 b) 3 c) 1 d) 5
A: 2

43. Q: What is the term for multiple processes executing concurrently in a single-user system?
a) Multiprocessing b) Multitasking c) Multithreading d) Time-sharing
A: Multitasking

44. Q: What configuration option does the PPP (Point-to-Point Protocol) system have?
a) Encryption b) Compression c) Authentication d) Multiplexing
A: Authentication

45. Q: How many articles are in the NEC code of conduct?
a) 5 b) 8 c) 10 d) 12
A: 8

46. Q: What is a key function of CMOS technology?
a) High speed b) Low power consumption c) High storage capacity d) High heat dissipation
A: Low power consumption

47. Q: What is single-level inheritance?
a) A class inherited from multiple base classes b) A class inherited from a base class c) A base class inherited from multiple derived classes d) A class that cannot be inherited
A: A class inherited from a base class

48. Q: What software model represents behavior and interaction?
a) Class diagram b) Sequence diagram c) Use case d) Activity diagram
A: Use case

49. Q: In which software development model is risk management prominently featured?
a) Agile b) Iterative c) Spiral d) Waterfall
A: Spiral model

50. Q: Which software development model is difficult to maintain?
a) Agile b) Iterative c) Spiral d) Waterfall
A: Waterfall

51. Q: What type of image format uses a map of bits?
a) Vector b) Bitmap c) JPEG d) GIF
A: Bitmap

52. Q: What technique is used to check if a language is regular?
a) Turing test b) Pumping lemma c) Halting problem d) Church-Turing thesis
A: Pumping lemma

53. Q: Which C function is used to find the current position in a file?
a) fseek() b) ftell() c) fgetpos() d) rewind()
A: ftell()

54. Q: Which SQL command removes content without altering the table structure?
a) DROP b) TRUNCATE c) DELETE d) REMOVE
A: DELETE

55. Q: What does software quality management consist of?
SQA and SQC b) SQA and SQM c) SQM and SQC d) SQP and SQA
A: SQA (Software Quality Assurance) and SQM (Software Quality Management)

56. Q: In a project network diagram, what is a dummy activity?
a) Critical activity b) Non-critical activity c) Theoretical or logical d) Resource-intensive activity
A: Theoretical or logical

57. Q: What is a limitation of Artificial Neural Networks?
a) Cannot learn from data b) Cannot handle complex problems c) Cannot explain results d) Cannot be trained
A: Cannot explain results

58. Q: Which of the following is not a physical input device?
a) Keyboard b) Mouse c) Touch panel d) Microphone
A: Touch panel

59. Q: Which of the following is not a property of knowledge representation?
a) Representational adequacy b) Inferential adequacy c) Inferential efficiency d) Representational verification
A: Representational verification

60. Which of the following is considered an Essential Implicant in Boolean algebra?

a) Prime implicant b) Essential prime implicant c) Complement

A: Essential prime implicant

Section B (20*2 = 40)

1) Q: What types of machine learning tasks are speech recognition and movie rating prediction, respectively?
   a) Classification and regression
   b) Regression and classification
   c) Clustering and regression
   d) Classification and clustering
   A: a) Classification and regression

2) Q: In C++, what is the output of the following code snippet?
   a) A b) B c) AB d) no output

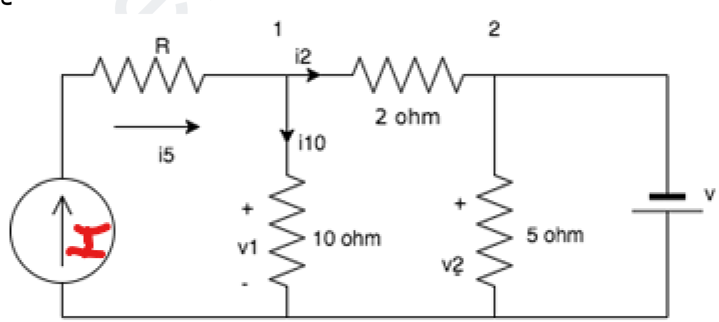
```
class A {
public:
    virtual void display() {
        cout << "A";
    }
};

class B : public A {
public:
    void display() override {
        cout << "B";
    }
};

int main() {
    A* ptr = new B();
    ptr->display();
    delete ptr;
    return 0;
}
```

A: b) B

3) Q: What is the value of the current I when v= 12V and v1=20V?



   a) 6A b)8A c) 2A d) 4A
   A: a)
   Explanation:

The current through the 10 ohm resistor = v1/10 = 2A.
Applying KCL at node 1: i5=i10+i2
i2=i5-2
Drop in 2 ohm resistor = 2 * i2
V1 = 20V; v2 = 20- (2*i2) => v=20-(2*i2) [v2=v as in parallel]
=> 12= 20- (2*i2) => i2 = 4 A
Thus, I= i5 = i2+2= 6A

4) Q: Given the production rules: Rule 1: S -> aSb and S -> e Rule 2: R -> cRd and R -> e How many production rules start with different alphabets in S union R?
    a) 6A b)8A c) 2A d) 4A

   A: 2

5) Q: What is the correct order of precedence for logical operators?
    a) Negation, AND, OR, Implication, Bidirectional
    b) AND, Negation, OR, Implication, Bidirectional
    c) OR, AND, NEGATION, Implication, Bidirectional
    d) Bidirectional, AND, OR, Implication, Negation

   A: a) Negation, AND, OR, Implication, Bidirectional

6) Q: In a half adder, which logic gate represents the carry?
    a) OR b)AND c) NANDd) NOT

   A: b) AND

7) In the demand paging memory, a page table is held in registers. If it takes 1000 ms to service a page fault and if the memory access time is 10 ms, what is the effective access time for a page fault rate of 0.01?
    a) 12.9 b)20.9 c) 19.9 d) 0.01

   A: 19.9ms

Data:

Page fault rate = p = 0.01,

Page service time = S = 1000 ms

Memory access time = T = 10 ms

Concept:

Effective access time = $(1 - p) \times T + p \times S$

Calculation:

Effective access time = $(1 - 0.01) \times 10 + 0.01 \times 1000$

Effective access time = $0.99 \times 10 + 10 = 9.9 + 10 = 19.9$ ms

The effective access time is 19.9 ms

8) A company purchases a piece of equipment for $10,000. The equipment has a useful life of 5 years with no salvage value at the end of its useful life. Calculate the annual depreciation percentage.
   a) 9 % b)15% c) 10% d) 20%
   A: d) 20%

1. **Calculate the Annual Depreciation Expense:**

   Using the straight-line method, the annual depreciation expense is calculated as:

$$\text{Annual Depreciation Expense} = \frac{\text{Cost of the Asset}}{\text{Useful Life}}$$

   Given:

   - Cost of the Asset = $10,000
   - Useful Life = 5 years

$$\text{Annual Depreciation Expense} = \frac{10,000}{5} = \$2,000$$

2. **Calculate the Depreciation Percentage:**

   The depreciation percentage per year is calculated as:

$$\text{Depreciation Percentage} = \left(\frac{\text{Annual Depreciation Expense}}{\text{Cost of the Asset}}\right) \times 100$$

   Using the values from above:

$$\text{Depreciation Percentage} = \left(\frac{2,000}{10,000}\right) \times 100 = 20\%$$

9) Q: What does the C function call fwrite(str, strlen(str) + 1, 1, filePointer) do?
a) Prints all string characters including the null character
b) Prints all string characters except the null character
c) Writes the length of the string to the file
d) Writes the string length plus one to the file

A: a) Prints all string characters including the null character

10) Q: What do building codes and bylaws primarily represent?

a) Design standards
b) Construction materials
c) Environmental regulations
d) Safety procedures

A: a) Design standards

11) Q: For a 4M-bit chip with 19 external connectors and 8-bit data lines, how many address lines are there?
    a) 8
    b) 16
    c) 19
    d) 20
    A: a) 8
    Explanation:

    Chip organization: 4M-bit
    - Total external connections: 19
    - Data lines: 8
    - A 4M-bit chip has 2^22 bits (since 4M = 4 * 1024 * 1024 = 2^22) - The number of address lines required to access 2^22 bits is 22. - Out of the 19 external connections, 8 are data lines, which leaves 11 for address lines and other control signals. - Since we already know that 8 data lines require 3 bits to represent 2^3 = 8 possible values, we can subtract 3 from the 11 remaining connections to get the number of address lines required. - Therefore, the number of address lines required = 11 - 3 = 8

12) Q: What is the result of 64 mod 23?
    a) 18
    b) 20
    c) 21
    d) 23

A: a) 18

13) Q: In a class B amplifier, what happens to voltage gain and impedance when capacitance is added to the emitter terminal?
    a) Both voltage gain and impedance increase
    b) Voltage gain increases, impedance decreases
    c) Voltage gain decreases, impedance increases
    d) Both voltage gain and impedance decrease

A: a) Both voltage gain and impedance increase

14) Q: In a tree traversal problem, which node is visited last in postorder when converting from preorder 30, 20, 10, 15, 25, 23, 39, 35, 42? (Problem Provided)
a) 10
b) 15
c) 23
d) 30

A: d) 30

EXPLANATION:

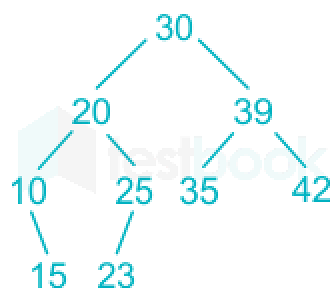
The **pre-order traversal** of given BST is:

30, 20, 10, 15, 25, 23, 39, 35, 42.

So, the **In-order traversal** of the BST is:

10, 15, 20, 23, 25, 30, 35, 39, 42.

The **Binary Search Tree** is:



So the post-order traversal of the tree is:

15, 10, 23, 25, 20, 35, 42, 39, 30

Hence, the correct answer is "option 4".

15) Q: In which region of a positively biased circuit does the Q-point typically lie?
a) Saturation
b) Active
c) Cut-off
d) Center

A: d) Center

16) Q: What are the three main phases of object-oriented development?
a) Analysis, Design, Testing
b) Object-oriented analysis, object-oriented design, object-oriented programming
c) Requirement gathering, Implementation, Deployment
d) Planning, Execution, Maintenance

A: b) Object-oriented analysis, object-oriented design, object-oriented programming

17) Q: What are the key components of a search problem in artificial intelligence?
a) Initial state, successor function, goal state, path cost
b) Initial state, successor function, goal path, goal test
c) Initial state, actions, goal state, search space
d) Initial state, goal state, heuristic function, solution

A: b) Initial state, successor function, goal path, goal test

18) Q: What is the default access specifier for data members in a C++ class?
a) Public
b) Protected
c) Private
d) None

A: c) Private

19) Q: What concept in object-oriented programming allows objects of different classes to be treated as objects of a common base class?
a) Encapsulation
b) Inheritance
c) Polymorphism
d) Abstraction

A: c) Polymorphism

20) Q: Which is not a property of representation of knowledge?
a) Representational Verification
b) Completeness
c) Consistency
d) Efficiency

A: a) Representational Verification

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Section A (60*1 = 100)

1. Q: What is the formula for no-load voltage gain in a common emitter configuration?
   a) Av = βRC/re
   b) Av = -βRC/re
   c) Av = RC/re
   d) Av = -RC/re
   A: d) Av = -RC/re

2. Q: What are the types of perspective projection?
   a) One-point and two-point perspective
   b) Two-point and three-point perspective
   c) One-point, two-point, and three-point perspective
   d) Only one-point perspective

A: c) One-point, two-point, and three-point perspective

3. Q: In a semaphore problem, what does "18P, xV, ..." represent?
   a) 18 signal operations and x wait operations
   b) 18 wait operations and x signal operations
   c) 18 processes and x variables
   d) 18 producers and x consumers
   A: b) 18 wait operations and x signal operations

4. Q: What is the latest amendment to the Labor Act?
   a) 2072
   b) 2073
   c) 2074
   d) 2075
   A: c) 2074

5. Q: How many parallel TCP connections does FTP typically use?
   a) 1
   b) 2
   c) 3
   d) 4
   A: b) 2

6. Q: Which of the following is based on the principle of conservation of charge?
   a) KCL
   b) KVL
   c) Ohm's Law
   d) Coulomb's Law
   A: a) KCL

7. Q: What is the primary purpose of a Project Charter?
   a) To outline the project scope and objectives
   b) To define the project schedule
   c) To identify project risks
   d) To allocate project resources
   A: a) To outline the project scope and objectives

8. Q: How many transistors are used in Class A amplifiers?
   a) 1
   b) 2
   c) 3
   d) 4
   A: a) 1

9. Q: What is the purpose of Gradient Descent?
   a) To maximize a function
   b) To minimize a function by iteratively moving in the direction of steepest descent
   c) To find the global maximum of a function

d) To solve linear equations
A: b) To minimize a function by iteratively moving in the direction of steepest descent

10. Q: The Waterfall model is the oldest approach for what purpose?
a) Software engineering
b) Software planning
c) Hardware design
d) Project management
A: a) Software engineering

11. Q: What is the IEEE standard for wireless networks?
a) 801.11
b) 802.11
c) 803.11
d) 804.11
A: b) 802.11

12. Q: What is Fan-out in digital circuits?
a) Number of inputs connected without degrading voltage
b) Number of standard loads that the output of a logic gate can drive
c) The speed at which a logic gate can operate
d) The power consumption of a logic gate
A: b) Number of standard loads that the output of a logic gate can drive

13. Q: What is the total number of registered professional engineers in NEC?
a) 51
b) 61
c) 71
d) 81
A: b) 61

14. Q: What is a viewport?
a) A virtual machine
b) A type of computer monitor
c) Visible area of a web page in a browser window
d) A graphics card component
A: c) Visible area of a web page in a browser window

15. Q: What is a feature of the 8086 microprocessor's internal architecture?
a) 8-bit data bus
b) 16-bit data bus
c) 32-bit data bus
d) 64-bit data bus
A: b) 16-bit data bus

16. Q: What does GUI stand for?
a) General User Interface
b) Graphical User Interface

c) Global User Interaction
d) Guided User Input
A: b) Graphical User Interface

17. Q: What does an IP address contain?
a) Only network portion
b) Only host portion
c) Network and host portions
d) MAC address
A: c) Network and host portions

18. Q: What type of value is a 230V rating on a heater?
a) Peak value
b) Average value
c) RMS (Root Mean Square) value
d) Instantaneous value
A: c) RMS (Root Mean Square) value

19. Q: Which function is typically used to classify into 3 or more classes?
a) Sigmoid function
b) ReLU function
c) Softmax function
d) Tanh function
A: c) Softmax function

20. Q: What is a contingent project?
a) A project with uncertain funding
b) A project that depends on the outcome of another project
c) A project with high risk
d) A project with multiple stakeholders
A: b) A project that depends on the outcome of another project

21. Q: In dimensionality, what does the symbol R typically refer to?
a) Rational numbers
b) Real numbers
c) Relative numbers
d) Random numbers
A: b) Real numbers

22. Q: Which is considered the fastest line drawing algorithm?
a) DDA algorithm
b) Bresenham's algorithm
c) Midpoint algorithm
d) Xiaolin Wu's algorithm
A: b) Bresenham's algorithm

23. Q: What shape does a unit square become after shearing?
a) Rectangle

b) Parallelogram
c) Rhombus
d) Trapezoid
A: b) Parallelogram

24. Q: What is the worst-case time complexity of Shell sort?
a) O(n)
b) O(n log n)
c) O(n^2)
d) O(n^3)
A: c) O(n^2)

25. Q: What is the tenure of the registrar of NEC (Nepal Engineering Council)?
a) 2 years
b) 3 years
c) 4 years
d) 5 years
A: c) 4 years

26. Q: What is the objective of NEA?
a) To regulate engineering practices in Nepal
b) To promote development of engineering science and technology in Nepal
c) To provide engineering education in Nepal
d) To certify engineering projects in Nepal
A: b) To promote development of engineering science and technology in Nepal

27. Q: Which of the following is not an application layer protocol?
a) FTP
b) SMTP
c) HTTP
d) TCP
A: d) TCP

28. Q: How many resonance frequencies does a crystal typically have?
a) 1
b) 2
c) 3
d) 4
A: b) 2

29. Q: How many flip-flops are required for representing flags of 8085?
a) 5
b) 6
c) 8
d) 10
A: a) 5

30. Q: Warshall algorithm gives:

a) Transitive closure
b) Shortest distance
c) Minimum spanning tree
d) Topological sorting
A: a) Transitive closure

31. Q: Which bias configuration is most stable?
a) Voltage Divider
b) CE
c) CB
d) CC
A: a) Voltage Divider

32. Q: SR flip flop is called:
a) Monostable
b) Bistable multivibrator
c) Astable multivibrator
d) Schmitt trigger
A: b) Bistable multivibrator

33. Q: Iterative Deepening DFS space complexity is:
a) b^d/2
b) O(bd)
c) O(d)
d) O(n)
A: c) O(d)

34. Q: ANN doesn't have:
a) Explanation of results
b) Learning capability
c) Parallel processing
d) Fault tolerance
A: a) Explanation of results

35. Q: What is a key property of bilateral circuits?
a) They only allow current to flow in one direction
b) They have different impedances for different directions of current flow
c) They have the same impedance regardless of the direction of current flow
d) They require AC power sources to function
A: c) They have the same impedance regardless of the direction of current flow

36. Q: Signed int are present in which C++ library?
a) std_1164
b) std_arth
c) cstdint
d) iostream
A: c) cstdint

37. Q: C++ program for loop syntax is:
   a) for (initialization; condition; increment/decrement)
   b) for (condition; initialization; increment/decrement)
   c) for (increment/decrement; condition; initialization)
   d) for (condition; increment/decrement; initialization)
   A: a) for (initialization; condition; increment/decrement)

38. Q: In a program, int *p = NULL means:
   a) p is a null pointer
   b) invalid assignment
   c) p is an integer
   d) p is a void pointer
   A: a) p is a null pointer

39. Q: If a derived class has a constructor, which of the following is true?
   a) Constructor of derived is called first and then that of base class
   b) Constructor of base class is called first and then that of derived class
   c) Only the derived class constructor is called
   d) Only the base class constructor is called
   A: b) Constructor of base class is called first and then that of derived class

40. Q: Pumping Lemma is used to detect:
   a) regular language
   b) non regular language
   c) CFG
   d) recursively enumerable language
   A: b) non regular language

41. Q: Non maskable interrupt is:
   a) TRAP
   b) INTR
   c) RST 7.5
   d) HOLD
   A: a) TRAP

42. Q: In Sequence diagram, vertical line indicates:
   a) time
   b) message
   c) object
   d) class
   A: a) time

43. Q: 8085 register connected to data bus:
   a) IR
   b) MBR
   c) PC
   d) MAR
   A: b) MBR

44. Q: IPSec which mode for privacy, integrity and authenticity:
   a) tunnel mode
   b) transport mode
   c) hybrid mode
   d) encapsulation mode
   A: a) tunnel mode

45. Q: If an SMTP server sends mail to another server, it is:
   a) SMTP client
   b) SMTP server
   c) POP3 client
   d) IMAP client
   A: a) SMTP client

46. Q: Connection less protocol:
   a) UDP
   b) TCP
   c) HTTP
   d) FTP
   A: a) UDP

47. Q: Finite state machine has how many tuples?
   a) 4
   b) 5
   c) 6
   d) 7
   A: c) 6

48. Q: Representation of composition of 2 TM languages:
   a) (TM1) u (TM2)
   b) TM1 + TM2
   c) TM1 * TM2
   d) TM1 ^ TM2
   A: b) TM1 + TM2

49. Q: 2D transformations require:
   a) 1D plane
   b) 2D plane
   c) 3D plane
   d) 4D plane
   A: b) 2D plane

50. Q: Binary tree of h height has at most ... nodes:
   a) 2^h - 1
   b) 2^(h+1) - 1
   c) 2^h
   d) 2^(h-1)

A: b) 2^(h+1) - 1

51. Q: Direct Access is seen in:
   a) disk
   b) RAM
   c) cache
   d) register
   A: a) disk

52. Q: Incremental model is:
   a) linear + waterfall
   b) linear + RAD
   c) iterative + waterfall
   d) iterative + RAD
   A: a) linear + waterfall

53. Q: Optimization algorithms use:
   a) heuristics
   b) statistics
   c) deterministic methods
   d) brute force
   A: a) heuristics

54. Q: Process of developing modules from sub systems:
   a) modular decomposition
   b) modular composition
   c) functional decomposition
   d) structural composition
   A: a) modular decomposition

55. Q: Which UML diagram is used to model the flow of control or data in a system?
   a) UML Diagram
   b) Activity Diagram
   c) Sequence Diagram
   d) Class Diagram
   A: b) Activity Diagram

56. Q: What does the association represent in UML?
   a) Relationship between classes
   b) Inheritance between classes
   c) Dependency between objects
   d) Encapsulation of data
   A: a) Relationship between classes

57. Q: How many tuples does a Turing machine have?
   a) 5
   b) 6
   c) 7

d) 8
A: c) 7

58. Q: To whom does the registrar submit annual plans and programs of council?
a) Government
b) PM
c) NEC
d) President
A: The question doesn't provide a clear answer.

59. Q: Port used for HTML:
a) 80
b) 443
c) 21
d) 25
A: a) 80

60. Q: What is taken for repetitive analysis?
a) LCM
b) HCF
c) Mean
d) Mode
A: a) LCM

Section B (20*2 = 40)

1. Q: In an alpha-beta pruning problem, what are the values of alpha and beta?
a) Fixed values of 0 and 1
b) Always positive integers
c) Always negative integers
d) Dynamic values that change during the search process

A: d) Dynamic values that change during the search process

2. Q: What is the correct order of processes in Computer Vision?
a) Image acquisition, segmentation, processing, analysis
b) Image acquisition, processing, segmentation, analysis
c) Image processing, acquisition, segmentation, analysis
d) Image segmentation, acquisition, processing, analysis

A: b) Image acquisition, processing, segmentation, analysis

3. Q: For a network with 10 nodes in a mesh topology, how many duplex connections are required?
   a) 45
   b) 90
   c) 100
   d) 10

   A: a) 45 (The formula for full mesh networks is n(n-1)/2, where n is the number of nodes. So, 10(10-1)/2 = 45)

4. Q: What is the result of a "wound and wait" problem?

T1	T2
R(A)	
	R(B)
	R(C)
W(B)	
	W(A)

a) T1 wounds T2 for B and T2 must wait for A.
b) T2 wounds T1 for A and T1 must wait for B.
c) T1 and T2 both proceed without any waits or wounds.
d) T2 wounds T1 for B and T1 must wait for A.

a) T1 wounds T2 for B and T2 must wait for A.

5. Q: In a given catch-throw program, what is the output?

```
#include <iostream>
  using namespace std;
  int main() {
     try {
        throw 20;
     }
     catch (int e) {
        cout << "Exception caught: " << e << endl;
```

```
    }
    return 0;
}
```

a) Exception caught: 0
b) Exception caught: 20
c) No exception caught
d) Runtime error
A: b) Exception caught: 20

6. Q: In a given namespace concept program, what should the next line of code be?

```
#include <iostream>
using namespace std;
namespace First {
    void sayHello() {
        cout << "Hello from First namespace!" << endl;
    }
}
namespace Second {
    void sayHello() {
        cout << "Hello from Second namespace!" << endl;
    }
}
int main() {
    First::sayHello();
    // Next line to be guessed:
```

a) Second.sayHello();
b) Second::sayHello();
c) sayHello();
d) ::sayHello();
A: b) Second::sayHello();

7. A 250V bulb passes a current of 0.3A. Calculate the power in the lamp.
a) 75W
b) 50W
c) 25W
d) 90W
A: a) 75W
Explanation: Here, V = 250v and I = 0.3A. P=VI. Which implies that, P=250*0.3=75W.

8. What type of relationship is represented when multiple orders are associated with a single customer?

A) Many-to-One
B) One-to-Many
C) Many-to-Many
D) One-to-One

Answer: A) Many-to-One

9. Find 2's complement of the binary number 10101101.

1. **Invert the digits (0s to 1s and 1s to 0s):**

   Original: 10101101

   Inverted: 01010010

2. **Add 1 to the least significant bit (LSB) of the inverted number:**

$$\begin{array}{cccccccc} 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ + & & & & & & & 1 \\ \hline 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \end{array}$$

So, the 2's complement of the binary number 10101101 is 01010011.

10. Which of the following strings can be derived from the given CFG?

a) aababb b) aaabbb c) ababab d) aabb

$$\begin{aligned} S &\rightarrow aS|Sb|AB|\epsilon \\ A &\rightarrow aA|a \\ B &\rightarrow bB|b \end{aligned}$$

A: a) aababb
Explanation:

1. Start with S→aS
2. S→aSb→aSb
3. Then, use S→AB
4. A→aA→aa
5. B→bB→bb

   So, S→aSb→aaSbb→aababb

11. Which of the following is a preemptive scheduling algorithm?

A) First-Come, First-Served (FCFS)

B) Shortest Job Next (SJN)

C) Round Robin (RR)

D) Highest Response Ratio Next (HRRN)

A: C) Round Robin (RR)

12. What is the formula for the sigmoid activation function?

A) $\sigma(x) = \frac{1}{1+e^{x}}$

B) $\sigma(x) = \frac{1}{1+e^{-x}}$

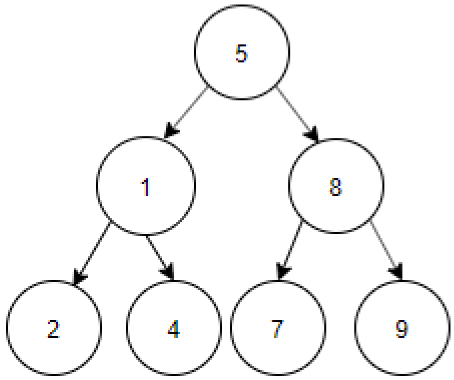
C) $\sigma(x) = e^{-x}$

D) $\sigma(x) = \frac{e^{x}}{1+e^{x}}$

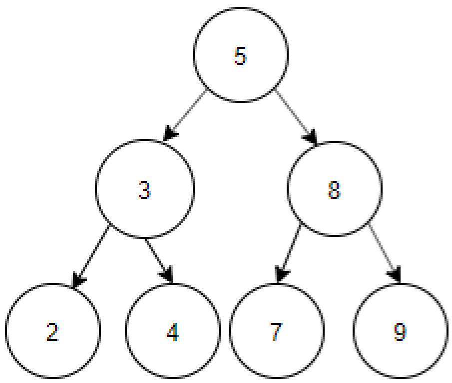
Answer: B) $\sigma(x) = \frac{1}{1+e^{-x}}$

13. Construct a binary search tree by using the postorder sequence given below.

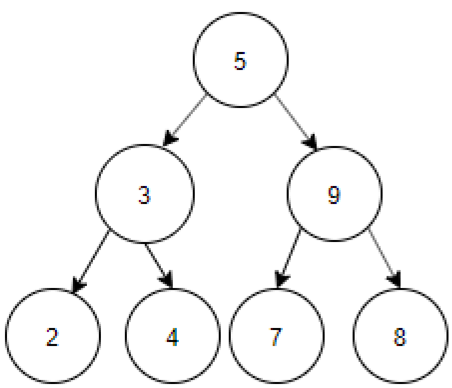
Postorder: 2, 4, 3, 7, 9, 8, 5.



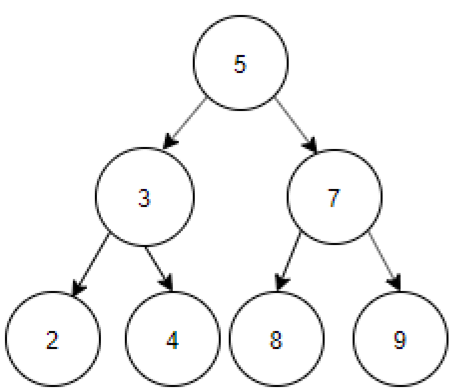
a)



b)



c)



d)

Answer: b

Explanation: Postorder sequence is 2, 4, 3, 7, 9, 8, 5.

Inorder sequence is the ascending order of nodes in Binary search tree. Thus, Inorder sequence is 2, 3, 4, 5, 7, 8, 9.

14. Determine the resonant frequency for the specifications: R = 10Ω, L = 0.1H, C = 10μF.

a) 157

b) 158

c) 159

d) 160

Answer: c

Explanation: The frequency at which the resonance occurs is called resonant frequency. The expression of the resonant frequency is given by fr = 1/(2π√LC). On substituting the given values we get resonant frequency = 1/(2π√(0.1×10×10-6))=159.2 Hz.

15. For an ideal operational amplifier in a closed-loop configuration with negative feedback, which of the following statements is true?

A) The input impedance is zero, and the output impedance is infinite.
B) The output voltage is determined by the input impedance and the load resistance.
C) The voltage difference between the inverting and non-inverting inputs is zero.
D) The current flowing into the inverting input is equal to the current flowing into the non-inverting input.

A: C) The voltage difference between the inverting and non-inverting inputs is zero.

16. What does overfitting refer to in machine learning?

A) The model performs well on new, unseen data but poorly on training data.

B) The model performs well on both training and new data.

C) The model performs well on training data but poorly on new, unseen data.

D) The model has a high bias and low variance.

Answer: C) The model performs well on training data but poorly on new, unseen data.

17. If a cache contains 16 words, and each word is 32 bits, what is the total cache size in bytes?

A) 32 bytes
B) 64 bytes
C) 128 bytes
D) 256 bytes

A: B) 64 bytes

Explanation:

The total cache size in bits is 16 words × 32 bits/word= 512 bits

16 words×32 bits/word=512 bits. Converting to bytes: 512 bits ÷ 8 bits/byte = 64 bytes

512 bits÷8 bits/byte=64 bytes.

18. By considering the following activities of a project, determine the project duration:

Activity	F	G	H	I	J
Immediate predecessors	-	F	F	G, H	G, I
Duration (days)	6	4	5	3	7

a) 13 days b) 16 days c) 20 days d) 25 days

Explanation : To determine the project duration, we need to identify the critical path, which is the longest sequence of dependent activities.

1. Start with activity F as it has no predecessors.
2. Activities G and H can start after F is completed.
3. Activity I can start after both G and H are completed.
4. Activity J can start after G and I are completed.

Let's calculate the earliest finish times for each activity:

- F: 0 + 6 = 6
- G: 6 + 4 = 10
- H: 6 + 5 = 11
- I: max(10, 11) + 3 = 14
- J: max(10, 14) + 7 = 21

The critical path is F -> H -> I -> J, with a total duration of 6 + 5 + 3 + 7 = 21 days.

Therefore, the correct answer is c) 20 days.

19. A bank advertises a nominal annual interest rate of 8% compounded quarterly on a savings account. What is the effective annual interest rate?

a) 8.00%

b) 8.24%

c) 8.37%

d) 8.43%

A: b)

Explanation: To find the effective annual interest rate, we use the formula:

Effective Annual Rate = (1 + r/n)^n - 1

20. What is correct about NAND Gates?

a) NAND gates are universal gates and can be used to construct all other basic logic gates.

b) NAND gates have the unique property of being able to represent both AND and OR operations simultaneously.

c) The Boolean algebra underlying NAND operations is more expressive than that of other logic gates.

d) NAND gates are immune to noise and signal degradation, allowing for more complex circuit designs.

A: a)

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Section A (60*1 = 60)

1. Q: What is it called when multiple threads are in execution?
Options:
a) Multiprocessing
b) Multithreading
c) Parallel processing
d) Concurrent execution
A: b) Multithreading

2. Q: Which two main processes does Natural Language Processing (NLP) contain?
Options:
a) Natural Language Understanding (NLU) and Natural Language Generation (NLG)
b) Natural Language Interpretation (NLI) and Natural Language Synthesis (NLS)
c) Natural Language Analysis (NLA) and Natural Language Production (NLP)
d) Natural Language Recognition (NLR) and Natural Language Output (NLO)
A: a) Natural Language Understanding (NLU) and Natural Language Generation (NLG)

3. Q: What is a key limitation of Finite State Machines (FSM)?
Options:
a) Cannot handle parallel processes
b) Cannot handle recursive functions
c) Cannot handle context-dependent behavior
d) Cannot handle non-deterministic input
A: c) Cannot handle context-dependent behavior

4. Q: How many parallel TCP connections does FTP use?
Options:
a) 1
b) 2
c) 3
d) 4
A: b) 2 (one for control and one for data)

5. Q: What is the formula for no-load voltage gain of a BJT in fixed bias configuration?
Options:
a) $-\beta(Rc/re)$
b) $\beta(Rc/re)$
c) $-(Rc/re)$
d) $(\beta/Rc)*re$
A: a) $-\beta(Rc/re)$, where β is current gain, Rc is collector resistance, re is emitter resistance

6. Q: What is the formula for 3D rotation around the z-axis?

a)

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$$

c)

$$\begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

d)

$$\begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

A: a)

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

7. Q: In a multiprocessor environment, how is synchronization typically achieved?
a) Using semaphores or mutexes
b) Using shared memory
c) Using cache coherence protocols
d) Using message passing
A: a)Using semaphores or mutexes

8. Q: What is the transformation matrix for oblique parallel projection?

a)

$$[T] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -f\cos\alpha & -f\sin\alpha & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

b)

$$\begin{bmatrix} 1 & 0 & a & 0 \\ 0 & 1 & b & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ a & b & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & a & 0 & 0 \\ b & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Answer: a)

9. Q: If three 2-ohm resistors are connected in a triangle, what is the resistance measured between any two terminals?
a) 0.67 ohms
b) 1 ohm
c) 1.33 ohms
d) 2 ohms
A: 1.33 ohms

10. Q: What is the time complexity of the quicksort algorithm in the average case?

a) O(n)
b) O(n^2)
c) O(n log n)
d) O(log n)
A: O(n log n)

11. Q: What is the purpose of virtual memory in operating systems?
a) To provide an illusion of larger main memory than physically available
b) To enhance the speed of the CPU
c) To increase the security of the system
d) To manage peripheral devices

A: a) To provide an illusion of larger main memory than physically available

12. Q: How many main types of parsing are there?
a) One: Top-down parsing
b) One: Bottom-up parsing
c) Two: Top-down parsing and Bottom-up parsing
d) Three: Top-down parsing, Bottom-up parsing, and Middle-out parsing

A: c) Two: Top-down parsing and Bottom-up parsing

13. Q: What is the difference between compiler and interpreter?
a) A compiler translates the entire program at once, while an interpreter translates line by line
b) A compiler translates line by line, while an interpreter translates the entire program at once
c) Both translate the entire program at once
d) Both translate line by line

A: a) A compiler translates the entire program at once, while an interpreter translates line by line

14. Q: How does capacitive reactance change with increasing frequency?
a) Capacitive reactance increases
b) Capacitive reactance decreases
c) Capacitive reactance remains constant
d) Capacitive reactance becomes zero

A: b) Capacitive reactance decreases with increasing frequency

15. Q: How many transistors are typically used in a Class A amplifier?
a) One transistor
b) Two transistors
c) Three transistors
d) Four transistors

A: a) One transistor

16. Q: What does Ti <A, v1, v2> typically represent in database systems?
a) A table
b) A transaction
c) An attribute
d) A view

A: b) A transaction

17. Q: If the annual worth is 9000, what is the capitalized worth at 10%?
a) 10,000
b) 45,000
c) 90,000
d) 100,000

A: c) 90,000

18. Q: What are the types of perspective projections?
a) Parallel projection, Orthographic projection, Isometric projection
b) One-point perspective, Two-point perspective, Three-point perspective
c) Linear perspective, Aerial perspective, Forced perspective
d) Vanishing point, Horizon line, Ground line

A: b) One-point perspective, Two-point perspective, Three-point perspective

19. Q: In a semaphore problem, what do P and V operations represent?
a) P represents the signal operation (increment), and V represents the wait operation (decrement)
b) P represents the wait operation (decrement), and V represents the signal operation (increment)
c) Both represent the wait operation
d) Both represent the signal operation

A: b) P represents the wait operation (decrement), and V represents the signal operation (increment)

20. Q: Who typically verifies and documents changes in a software project?
a) Project Manager
b) Software Quality Assurance (SQA) team
c) Development Team
d) End Users

A: b) Software Quality Assurance (SQA) team

21. Q: The Waterfall model is the oldest approach for what?

a) System engineering
b) Software testing
c) Software engineering
d) Project management

A: c) Software engineering

22. Q: What does "fan-in" refer to in digital logic?
a) The number of outputs a logic gate can handle
b) The power consumption of a logic gate
c) The speed of a logic gate
d) The number of inputs a logic gate can handle

A: d) The number of inputs a logic gate can handle

23. Q: What is admissibility in the context of the A* algorithm?
a) The heuristic overestimates the cost to the goal
b) The heuristic never underestimates the cost to the goal
c) The heuristic never overestimates the cost to the goal
d) The heuristic is always accurate

A: c) The heuristic never overestimates the cost to the goal

24. Q: Who is considered the "Father of AI"?
a) Alan Turing
b) John McCarthy
c) Marvin Minsky
d) Herbert Simon

A: b) John McCarthy

25. Q: What type of connector is typically used in Shielded Twisted Pair (STP) cable?
a) BNC connector
b) RJ11 connector
c) RJ45 connector
d) SC connector

A: c) RJ45 connector

26. Q: What is the purpose of the volatile keyword in C?
a) To optimize the variable for faster access
b) To indicate that a variable may change unexpectedly
c) To allocate memory dynamically
d) To protect the variable from modification

A: b) To indicate that a variable may change unexpectedly

27. Q: What is the difference between process and thread?

a) A process is an independent program, while a thread is a subset of a process
b) A thread is an independent program, while a process is a subset of a thread
c) A process runs within a thread, while a thread runs independently
d) A thread has its own memory space, while a process shares memory space

A: a) A process is an independent program, while a thread is a subset of a process

28. Q: What is the output of a given C++ program?
a) 0
b) 1
c)]
d) [

A: c)]

29. Q: What type of oscillator is a Wien bridge oscillator?
a) High-frequency oscillator
b) Low-frequency oscillator
c) Medium-frequency oscillator
d) Voltage-controlled oscillator

A: b) Low-frequency oscillator

30. Q: Which of the following is not a decoder: 4:2, 8:3, 16:4, or 5:32?
a) 4:2
b) 8:3
c) 16:4
d) 5:32

A: d) 5:32 is not a standard decoder configuration

31. Q: Which of these cannot be used as a variable name in C: true or while?
a) true
b) while

A: b) while (it's a reserved keyword in C)

32. Q: After which step is software typically ready in the spiral model?
a) Usually after the 1st iteration
b) Usually after the 2nd iteration
c) Usually after the 3rd iteration
d) Usually after the 4th or 5th iteration

A: d) Usually after the 4th or 5th iteration

33. Q: Priority interruptions make use of which mechanisms?
a) Interrupt masking and polling
b) Polling and daisy chain

c) Interrupt masking and daisy chain
d) Daisy chain and polling

A: d) Daisy chain and polling

34. Q: To which pin does the CPU send a signal as a response to the Programmable Interrupt Controller (PIC)?
a) INTA (Interrupt Acknowledge)
b) IRQ (Interrupt Request)
c) NMI (Non-Maskable Interrupt)
d) INTR (Interrupt Request)

A: a) INTA (Interrupt Acknowledge)

35. Q: What are the three main elements of a use case?
a) Actor, System, and Goal
b) Actor, System, and Process
c) System, Goal, and Action
d) Actor, Goal, and Process

A: a) Actor, System, and Goal

36. Q: What is the primary function of a Device Driver?
a) To provide an interface between the operating system and hardware devices
b) To manage system memory
c) To execute user programs
d) To handle network communication

A: a) To provide an interface between the operating system and hardware devices

37. Q: What are some unique features of RISC (Reduced Instruction Set Computer) architecture?
a) Complex instructions, fewer registers, and complex addressing modes
b) Simple instructions, fewer registers, and complex addressing modes
c) Simple instructions, more registers, and load-store architecture
d) Complex instructions, more registers, and load-store architecture

A: c) Simple instructions, more registers, and load-store architecture

38. Q: What happens when an exception occurs in C++?
a) The program terminates immediately
b) The function in which the exception occurs is ignored
c) A function (exception handler) is called
d) The program continues to execute without handling the exception

A: c) A function (exception handler) is called

39. Q: How can a weak entity set be changed into a strong entity set?

a) By introducing a foreign key
b) By introducing a primary key
c) By removing all its attributes
d) By merging it with another weak entity set

A: b) By introducing a primary key

40. Q: Which architecture does FTP use?
a) Peer-to-peer architecture
b) Client-server architecture
c) Distributed architecture
d) Multi-tier architecture

A: b) Client-server architecture

41. Q: Which is considered the most important step in a Genetic Algorithm? Options: a) Mutation b) Crossover c) Selection d) Initialization A: c) Selection

42. Q: Which type of cryptography uses two keys? Options: a) Symmetric cryptography b) Asymmetric cryptography c) Hash-based cryptography d) Elliptic curve cryptography A: b) Asymmetric cryptography

43. Q: What is used for overloading in programming? Options: a) Only type b) Only arguments c) Both type and arguments d) Neither type nor arguments A: c) Both type and arguments

44. Q: Which of these is a valid definition of a pointer in C? Options: a) p = *a b) p = &a c) *p = a d) &p = a A: b) p = &a

45. Q: A tautology is a statement which is always what? Options: a) False b) True c) Uncertain d) Contradictory A: b) True

46. Q: Excess-3 code is also known as? Options: a) Self-correcting code b) Self-completing code c) Self-adjusting code d) Self-encrypting code A: b) Self-completing code

47. Q: What is the purpose of normalization in database design? Options: a) To increase data redundancy b) To reduce data integrity c) To reduce data redundancy and improve data integrity d) To increase database size A: c) To reduce data redundancy and improve data integrity

48. Q: What is the difference between Stack and Queue data structures? Options: a) Stack is FIFO, Queue is LIFO b) Stack is LIFO, Queue is FIFO c) Both are FIFO d) Both are LIFO A: b) Stack follows Last-In-First-Out (LIFO), while Queue follows First-In-First-Out (FIFO)

49. Q: What is the purpose of a virtual function in C++? Options: a) To achieve compile-time polymorphism b) To achieve runtime polymorphism c) To improve memory efficiency d) To prevent function overloading A: b) To achieve runtime polymorphism

50. Q: What is the main advantage of using optical fiber for communication? Options: a) Low cost b) Easy installation c) High bandwidth and low signal loss d) Compatibility with older systems A: c) High bandwidth and low signal loss

51. Q: What is the purpose of the OSI model in networking?
Options:
a) To increase network speed
b) To standardize the communication functions of a telecommunication system
c) To reduce hardware costs
d) To encrypt network traffic
A: b) To standardize the communication functions of a telecommunication system

52. Q: What is the difference between static and dynamic binding in programming?
Options:
a) Static binding is slower, dynamic binding is faster
b) Static binding uses more memory, dynamic binding uses less
c) Static binding occurs at compile-time, while dynamic binding occurs at run-time
d) Static binding is for object-oriented languages, dynamic binding is for procedural languages
A: c) Static binding occurs at compile-time, while dynamic binding occurs at run-time

53. Q: What is the purpose of a cache in computer architecture?
Options:
a) To increase the main memory size
b) To reduce the average time to access data from the main memory
c) To store permanent data
d) To improve CPU clock speed
A: b) To reduce the average time to access data from the main memory

54. Q: What is the difference between cohesion and coupling in software engineering?
Options:
a) Cohesion is about modules, coupling is about data
b) Cohesion is internal, coupling is external
c) Cohesion measures relationships within a module, coupling measures interdependence between modules
d) Cohesion is for design, coupling is for implementation
A: c) Cohesion measures the strength of relationships within a module, while coupling measures the interdependence between modules

55. Q: What is the purpose of the volatile keyword in Java?
Options:
a) To make a variable constant
b) To indicate that a variable's value can be changed by multiple threads simultaneously
c) To improve variable access speed
d) To prevent variable initialization
A: b) To indicate that a variable's value can be changed by multiple threads simultaneously

56. Q: What is the difference between TCP and UDP protocols?
Options:
a) TCP is faster, UDP is more reliable
b) TCP uses IP, UDP doesn't
c) TCP is connection-oriented and reliable, UDP is connectionless and unreliable but faster
d) TCP is for web browsing, UDP is for email
A: c) TCP is connection-oriented and reliable, while UDP is connectionless and unreliable but faster

57. Q: What is the purpose of a Fourier transform in signal processing?
Options:
a) To amplify signals
b) To filter noise
c) To convert a signal from the time domain to the frequency domain
d) To compress data
A: c) To convert a signal from the time domain to the frequency domain

58. Q: What is the difference between deep copy and shallow copy in object-oriented programming?
Options:
a) Deep copy is faster, shallow copy is slower
b) Deep copy uses less memory, shallow copy uses more
c) Deep copy creates a new object and recursively copies nested objects, shallow copy copies references
d) Deep copy is for primitive types, shallow copy is for objects
A: c) Deep copy creates a new object and recursively copies nested objects, while shallow copy creates a new object but copies references to nested objects

59. Q: What is the purpose of a load balancer in network architecture?
Options:
a) To increase network security
b) To distribute incoming network traffic across multiple servers
c) To compress network data
d) To cache frequently accessed data
A: b) To distribute incoming network traffic across multiple servers to ensure no single server becomes overwhelmed

60. Q: What is the difference between supervised and unsupervised learning in machine learning?
Options:
a) Supervised learning is automatic, unsupervised learning requires human intervention
b) Supervised learning uses labeled data, unsupervised learning uses unlabeled data
c) Supervised learning is for classification, unsupervised learning is for regression
d) Supervised learning is slower, unsupervised learning is faster
A: b) Supervised learning uses labeled data for training, while unsupervised learning uses unlabeled data

Section 2 (20*2=40)

1. Q: Which of the following expressions represents the logical AND operation followed by the logical OR operation in sequence?

   A) $T_{ij}(k) = T_{ij}(k-1) \text{ AND } T_{ij}(k-1) \text{ OR } T_{ij}(k-1)$
   B) $T_{ij}(k) = T_{ij}(k-1) \text{ OR } (T_{ij}(k-1) \text{ AND } T_{ij}(k-1))$
   C) $T_{ij}(k) = (T_{ij}(k-1) \text{ AND } T_{ij}(k-1)) \text{ OR } T_{ij}(k-1)$
   D) $T_{ij}(k) = T_{ij}(k-1) \text{ AND } (T_{ij}(k-1) \text{ OR } T_{ij}(k-1))$

   A: B) Tij(k) = Tij(k-1) OR (Tij(k-1) AND Tij (k-1))

2. Which of the following terms best describes a problem for which there exists an algorithm that always halts with a correct yes/no answer?

   A) Turing Decidable
   B) Recursive
   C) Turing Undecidable
   D) Recursive Enumerable

   Answer: A) Turing Decidable

3. What is the primary purpose of decomposing a reference architecture?
   A) To increase the overall system's complexity
   B) To simplify the design and implementation process by breaking it into manageable components
   C) To eliminate the need for standardization
   D) To reduce the number of components in the system
   Answer: B) To simplify the design and implementation process by breaking it into manageable components

4. Which of the following correctly describes a one-to-many relationship?
   A) One department can have multiple employees.
   B) One student can enroll in multiple courses, and each course can have multiple students.
   C) Each order can be linked to multiple customers.
   D) Each customer can place multiple orders.

   Answer: A) One department can have multiple employees.

5. Given the vector (2,0,3)(2, 0, 3)(2,0,3) and the scaling vector (2,2,4)(2, 2, 4)(2,2,4), what is the result of scaling the original vector by the scaling vector?

   A) (4,0,12)(4, 0, 12)(4,0,12)
   B) (4,2,7)(4, 2, 7)(4,2,7)
   C) (2,0,7)(2, 0, 7)(2,0,7)
   D) (1,0,3)(1, 0, 3)(1,0,3)

   Answer: a) (4,0,12)(4, 0, 12)(4,0,12)

6. In a genetic algorithm used to generate class routines, if the fitness function is defined as f(x)= 1/ (1+ routine conflict) what does this fitness function measure?

   A) The total number of classes in the schedule
   B) The number of conflicts in the schedule
   C) The total duration of the class schedule
   D) The number of students enrolled in the classes

Answer: b) The number of conflicts in the schedule

7. In the demand paging memory, a page table is held in registers. If it takes 1000ms to service a page fault and if the memory access time is 20ms, what is the effective access time for a page fault rate of 0.01?
   a) 30.8 ms b) 40ms c) 22ms d) 29.8ms
   A:

Data:

Page fault rate = p = 0.01,

Page service time = S = 1000 ms

Memory access time = T = 20 ms

Concept:

Effective access time = p × T + (1 - p) × S

Calculation:

Effective access time = (1 - 0.01) × 20 + 0.01 × 1000

Effective access time = 0.99 × 20 + 10 = 19.8 + 10 = 29.8 ms

The effective access time is 29.8 ms

8. Capacitance is directly proportional to__________
a) Area of cross section between the plates
b) Distance of separation between the plates

c) Both area and distance
d) Neither area nor distance

A: a)

Explanation: The relation between capacitance, area and distance between the plates is: C=epsilon*A/D. According to this relation, the capacitance is directly proportional to the area.

9. Convert the decimal number 156 to its octal representation.
   a) 254 b)20 c)1234 d)234

A: d)

Explanation:

To convert the decimal number 156 to octal, you divide the number by 8 and keep track of the remainders.

1. 156÷8= 19 with a remainder of 4
2. 19÷8=2 with a remainder of 3
3. 2÷8=0 with a remainder of 2

Reading the remainders from bottom to top, the octal representation of 156 is 234.

10. Identify error in the program below;

```
#include <iostream>

using namespace std;

void printArray(int arr[], int size) {

  for(int i = 0; i <= size; i++) {

     cout << arr[i] << " ";

  }

  cout << endl;

}

int main() {

  int myArray[] = {1, 2, 3, 4, 5};

  int arraySize = sizeof(myArray) / sizeof(myArray[0]);
```

```
    printArray(myArray, arraySize);

    return 0;

}
```

A) Syntax Error
B) Runtime Error
C) Logical Error
D) Linker Error

A: B) Runtime Error

11. What will be the output in following C++ program.

```
int main() {

    int a = 5;

    int *p = &a;

    *p = *p * 2;

    a = a + 3;

    cout << *p << endl;

    cout << a << endl;

    return 0;

}
```

A)
10
13

B)
8
8

C)
10
10

D)
8
10

Answer: A)

12. What is output by following production rule?

$$
\begin{aligned}
S &\rightarrow AB \mid BA \\
A &\rightarrow aA \mid a \\
B &\rightarrow bB \mid b
\end{aligned}
$$

Answer: B) aabb

Explanation:

To derive the string ‘aabb’:

1. Start with S→AB
2. A→aA→aa
3. B→bB→bb
4. bB
5. bbB→bB→bb

So, S→AB→aabb

13. Which category of transmission media includes twisted pair cables?

a) Guided Media b) Coaxial cables c) Fiber optic cables d) Unguided Media
A: a) Guided Media

14. Consider the following four processes in the shortest job next scheduling. Calculate the average turnaround time.

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	2
P4	3	6

A) 10
B) 11.5

C) 12
D) 13

A: b) 11.5

Explanation:
P1 completes first from 0 to 8. Then the shortest job is of p3 and continues till 10. Then P2 runs from 10 to 14 and at last P4 runs from 14 to 20.

Calculating Turnaround Time:

- Turnaround Time = Completion Time - Arrival Time
- P1: Completion Time = 8, Arrival Time = 0
  - Turnaround Time = 8 - 0 = 8
- P2: Completion Time = 14, Arrival Time = 1
  - Turnaround Time = 14 - 1 = 13
- P3: Completion Time = 10, Arrival Time = 2
  - Turnaround Time = 10 - 2 = 8
- P4: Completion Time = 20, Arrival Time = 3
  - Turnaround Time = 20 - 3 = 17

Average Turnaround Time = (8 + 13 + 8 + 17) / 4 = 46 / 4 = 11.5

15. What is the correct order of phases in the Waterfall model of software development?

A) Requirement Analysis → System Design → Implementation → Integration and Testing → Deployment → Maintenance
B) Requirement Analysis → Implementation → System Design → Integration and Testing → Deployment → Maintenance
C) Implementation → System Design → Requirement Analysis → Integration and Testing → Deployment → Maintenance
D) System Design → Requirement Analysis → Implementation → Integration and Testing → Maintenance → Deployment

A: A) Requirement Analysis → System Design → Implementation → Integration and Testing → Deployment → Maintenance

16. Which of the following characteristics is true for an ideal operational amplifier?

A) The ideal op-amp has a finite open-loop gain, resulting in a non-zero voltage difference between the inverting and non-inverting inputs.
B) The ideal op-amp has infinite input impedance and zero output impedance.
C) The ideal op-amp can only be used in linear configurations and cannot be used in switching applications.
D) The ideal op-amp has a non-zero input bias current, which affects the accuracy of the amplifier.

A: B) The ideal op-amp has infinite input impedance and zero output impedance.

17. In supervised learning, what is the main objective during training?

A) To group similar data points
B) To predict outputs without labeled data
C) To learn the mapping from inputs to outputs using labeled data
D) To find hidden patterns without predefined labels

A: C) To learn the mapping from inputs to outputs using labeled data

18. In a full binary tree if the number of internal nodes is I, then the number of leaves L are?

a) L = 2*I

b) L = I + 1

c) L = I – 1

d) L = 2*I – 1

Answer: b)

Explanation: Number of Leaf nodes in full binary tree is equal to 1 + Number of Internal Nodes i.e L = I + 1

19. A process refers to 5 pages, A, B, C, D, E in the order : A, B, C, D, A, B, E, A, B, C, D, E. If the page replacement algorithm is FIFO, the number of page frames is increased to 4, then the number of page transfers ___________

a) decreases

b) increases

c) remains the same

d) none of the mentioned

A: b) increases

20. You plan to invest $1,000 in a savings account that offers an annual interest rate of 5%, compounded quarterly. You want to know the future value of this investment after 3 years.

a) $1130 b) $1227.5 c) $1500 d) $2000

Formula for Future Value with Compound Interest:

$$FV = P\left(1 + \frac{r}{n}\right)^{nt}$$

A: b)